

FLOOD BASALT The result of a massive eruption that covers an extensive area with basalt. See also *basalt*.

FLOODPLAIN A flat area alongside a river that is liable to be covered with water when river volumes are particularly great. See also *alluvium*.

FLOWSTONE A coating of a carbonate mineral such as calcite on the walls or floor of a cave. The mineral precipitates out of flowing water. See also *calcite*, *precipitate*.

FOG A cloud of water droplets (or ice crystals) suspended in the air close to Earth's surface. See also *radiation fog*.

FOLD See panel, below.

FOLIATION Repetitive layering in metamorphic rocks where minerals become arranged in parallel bands. See also *lamination*.

FORESHORE The part of a shore lying between the low-water level and the average high-water level. See also *beach* (panel), *beach face*.

FORKED LIGHTNING A cloud-to-ground electrical discharge with a branched path in a storm. See also *sheet lightning*, *thunderstorm*.

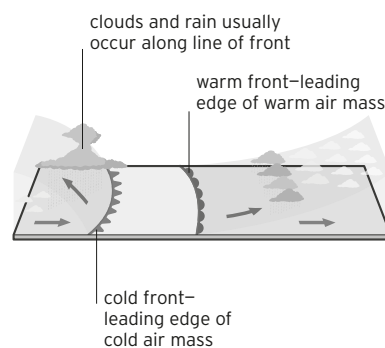
FOSSIL The impression or remains of an ancient animal or plant buried in rock and preserved in petrified form.

FOSSIL FUEL An energy source such as coal, natural gas, or oil that is derived from the remains of long-dead plants and animals buried beneath the ground.

FOSSILIZATION The process of being transformed into a fossil.

FRINGING REEF A coral reef where either there is no lagoon between the reef and the shore or, if there is, it is entirely shallow. See also *atoll*, *coral reef*.

FRONT ▼ In meteorology, the forward-moving edge of an air mass. See also *air mass*.



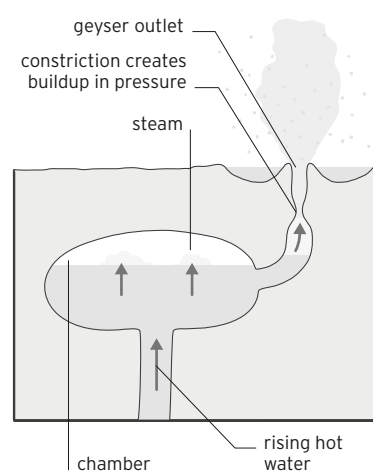
FUMAROLE A small opening in the ground in volcanic regions through which steam and hot gases escape.

FUNNEL A tube of rapidly rotating air descending from a cloud. If its lower end touches the ground, it becomes a tornado. See also *tornado*.

G

GEODE A hollow cavity in a rock that is lined with crystals.

GEO THERMAL Relating to the heat energy that is generated in Earth's interior. Geothermal energy is mainly produced by natural radioactive decay of uranium, thorium, and potassium.



GEYSER ▲ A jet of steam and boiling water that periodically erupts from the ground. Geysers are powered by hot rocks heating groundwater. See also *hot spring*.

GLACIER A slowly moving mass of ice formed by the accumulation and compaction of snow over a long period. There are many different types of glacier. See also *cirque glacier*, *piedmont glacier*, *surge-type glacier*, *valley glacier*.

GORGE A deep, narrow valley bounded by vertical or near-vertical cliffs. See also *canyon*.

GRABEN A block of Earth's crust lying between parallel faults and depressed relative to the blocks on either side. See also *rift valley*.

GRAIN The texture of a rock. For example, clay is made up of fine particles so is described as fine-grained, whereas conglomerate is coarse-grained.

GRANITE A coarse-grained igneous rock made up of the minerals quartz, feldspar, and mica.

GREASE ICE An early stage of sea-ice formation when tiny ice crystals called frazil join together to give the water surface the appearance of an oil slick. See also *sea-ice*.

GREENHOUSE EFFECT The tendency of the atmosphere to absorb a proportion of the Sun's radiation after it has been reradiated by Earth. As the amount of carbon dioxide in the atmosphere has increased, the greenhouse effect has become more marked. See also *greenhouse gas*.

GREENHOUSE GAS Any gas that promotes the greenhouse effect. The major greenhouse gases are carbon dioxide, methane, and water vapor. See also *greenhouse effect*.

FOLD

A geological structure in which once-horizontal layers of rock have been flexed and bent is called a fold. Folds develop when rocks are subjected to compression or tension. If the layers bend upward to form a ridge, the fold is an anticline; if they bend downward to form a trough, the

trough is a syncline. Folds can be symmetrical or asymmetrical. Overfolds and recumbents are asymmetrical folds where the layers have folded beyond 90°. In an overfold, the layers have folded back on themselves. A fold's axis is its angle of inclination to Earth's surface.

